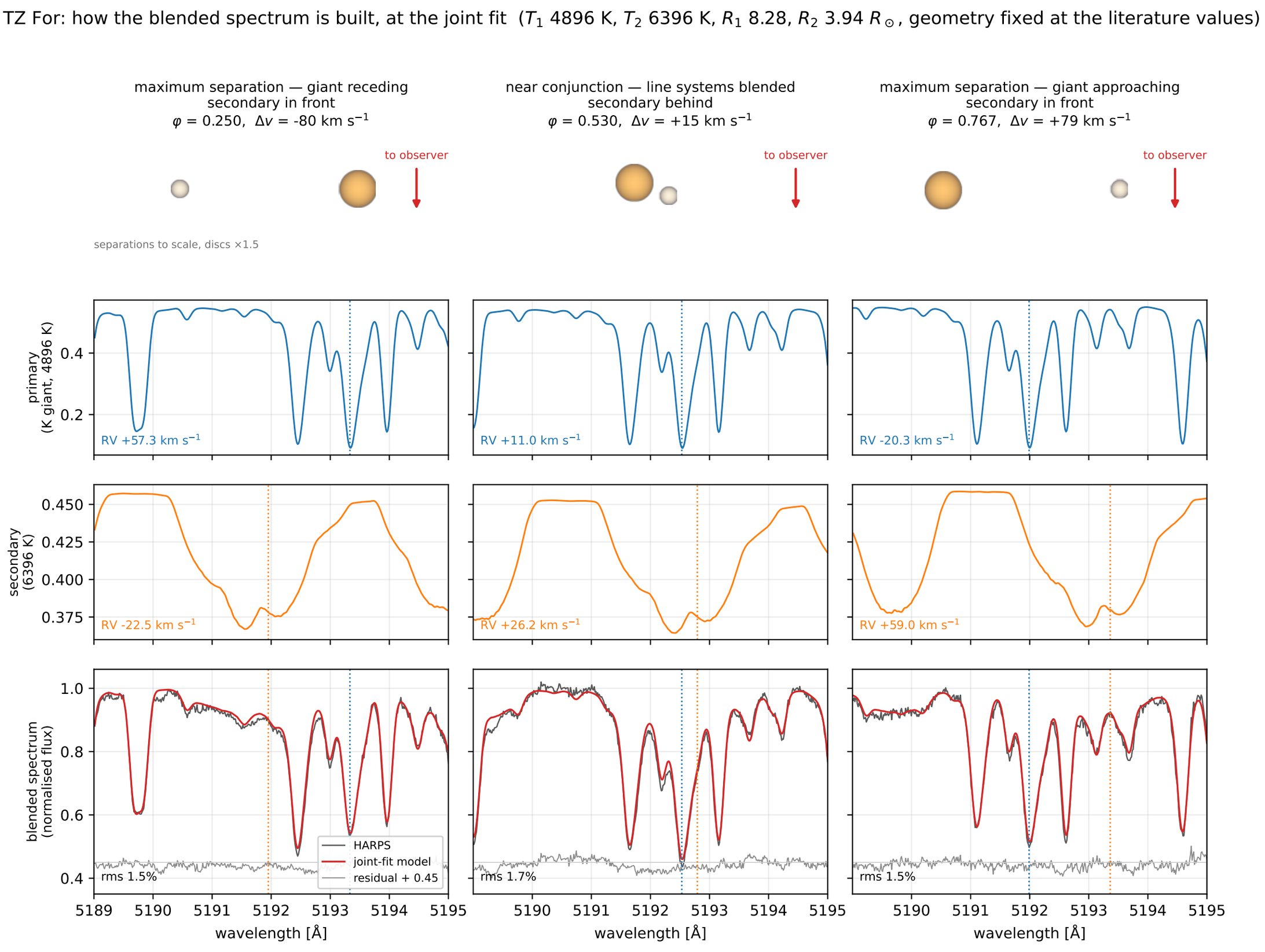

TZ For: how the blended spectrum is built, at the joint fit (T1 4896 K, T2 6396 K, R1 8.28, R2 3.94 R⊙, geometry fixed at the literature values)
maximum separation — giant receding
secondary in front
φ = 0.250, Δv = -80 km s−1
near conjunction — line systems blended
secondary behind
φ = 0.530, Δv = +15 km s−1
maximum separation — giant approaching
secondary in front
φ = 0.767, Δv = +79 km s−1
to observer
to observer
to observer
separations to scale, discs ×1.5
primary
(K giant, 4896 K)
RV +57.3 km s−1
RV +11.0 km s−1
RV -20.3 km s−1
secondary
(6396 K)
RV -22.5 km s−1
RV +26.2 km s−1
RV +59.0 km s−1
blended spectrum
(normalised flux)
HARPS
joint-fit model
residual + 0.45
rms 1.5%
rms 1.7%
rms 1.5%
wavelength [Å]
wavelength [Å]
wavelength [Å]